Network Switch Replacement & Installation Training Guide

Purpose

This document provides comprehensive, field-ready training for technicians performing **network switch replacement and installation**, including **extensive PuTTY usage** for console-based network access. It follows ITIL v4 Service Operation principles, Lean execution, and operational best practices.

---

1. Role of the Field Technician

A field technician is responsible for: - Restoring or improving service availability - Executing approved changes safely - Minimizing downtime and risk - Communicating clearly with stakeholders - Leaving the environment stable and documented

This task typically falls under: - **ITIL v4 – Service Operation** - **Change Enablement (Standard / Normal Change)** - **Incident Resolution (if replacement is reactive)**

---

2. Pre-Engagement Preparation

2.1 Understanding the Task Scope

Before attending site, confirm: - Is this a **planned replacement** or **fault-driven swap**? - Are configurations pre-staged or must be captured on-site? - Is downtime approved? - Who is the **Local Contact (LCON)**?

2.2 Required Tools & Equipment

Category	Items
Computing	Laptop, charger
Software	PuTTY, USB drivers
Connectivity	RJ45-USB console cable, USB-C console cable
Cabling	Ethernet cables (various lengths)
Power	Spare PSU (if modular), power leads
Access	Scissor lift / ladder (if required)
Diagnostics	Multimeter
Safety	PPE, gloves

---

3. On-Site Workflow Overview

1. Arrival & communication
2. Old switch assessment
3. Configuration capture
4. Physical removal
5. New switch installation
6. Console configuration
7. Network validation
8. Handover & documentation

Each step must be validated before moving to the next.

---

4. Old Switch Assessment & Preparation

4.1 Arrival Procedure

- Call LCON immediately
- Confirm permission to proceed
- Verify device identity (model, location)

4.2 Visual Inspection

Check: - Power LEDs - Status LEDs - Fan noise - Cable strain or damage

4.3 Configuration Capture (Critical)

Before disconnecting anything: - Console into the switch - Record: - Hostname - Interface status - VLAN configuration - Uplink ports - Take photos of front panel cabling

Failure to capture this data is a primary cause of extended outages.

---

5. Physical Removal Process

1. Shut down switch if instructed
2. Power off
3. Disconnect **access ports first**
4. Disconnect **uplinks last**
5. Remove PSU (if reusable)
6. Remove switch from rack or ceiling

Always support the device when removing to avoid cable or port damage.

---

6. New Switch Installation

6.1 Mounting

- Secure switch firmly
- Ensure airflow clearance
- Install PSU(s)

6.2 Initial Power-On

- Apply power
- Observe boot sequence
- Confirm no hardware alarms

Do not connect network cables yet.

---

7. PuTTY – Extensive Training

7.1 What is PuTTY?

PuTTY is a lightweight terminal emulator used to access devices via: - Serial (Console) - SSH - Telnet (legacy)

For switch replacement, **Serial Console** is most common.

7.2 Installing & Preparing PuTTY

1. Download PuTTY
2. Install USB-to-Serial drivers (if required)
3. Connect console cable
4. Open **Device Manager**
5. Identify assigned COM port

7.3 PuTTY Serial Configuration

Recommended settings: - Connection Type: **Serial** - Serial Line: COMx - Speed (Baud): 9600 (unless specified) - Data Bits: 8 - Stop Bits: 1 - Parity: None - Flow Control: None

Save sessions to avoid reconfiguration.

7.4 Navigating the Console

Common actions: - Press **Enter** to wake console - Use **Tab** for command completion - Use **?** to view available commands - Avoid copy/paste unless certain of syntax

---

7.5 Common Console Tasks

- Verify interfaces
- Check VLANs
- Confirm link status
- Review logs
- Assign management IP (if required)

Never factory reset unless explicitly instructed.

7.6 PuTTY Best Practices

- Always log sessions when possible
- Do not leave sessions unattended
- Disconnect safely before removing cable
- Label saved sessions clearly

---

8. Network Reconnection & Validation

8.1 Connection Order

1. Uplink ports first
2. Verify link and stability
3. Connect access ports gradually

8.2 Validation Checks

- Interfaces up/up
- No excessive errors
- VLANs correct
- End devices online

Allow 5–10 minutes of monitoring.

---

9. Handover & Closure

Before leaving: - Confirm service restoration with LCON - Provide status update - Document: - Serial numbers - Actions taken - Final state

Documentation protects both technician and organization.

---

10. Key Skills for Successful Technicians

Technical Skills

- Console access
- Basic networking (VLANs, uplinks)
- Power handling
- Cable management

Operational Skills

- Communication
- Change discipline
- Situational awareness
- Escalation judgement

---

11. Final Technician Principles

- Move deliberately
- Validate every step
- Communicate early
- Document always

A structured technician prevents incidents instead of creating them.

End of Training Guide